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Abstract

With the rapid integration of generative artificial intelligence into programming education, concerns have emerged regarding university students' potential dependency on such technologies. However, there is a lack of validated instruments to systematically measure generative AI Dependency on programming contexts. To address this gap, the present study aimed to develop and validate the Scale of Generative AI Dependency on Programming among University Students. The study further seeks to contribute to the conceptualization of GenAI dependency as a context-specific construct in programming education, rather than treating students' reliance on GenAI as a general or undifferentiated form of technology use. Drawing on prior research and theoretical frameworks related to technology dependency and learning behaviors, an initial item pool was constructed to capture students' dependency on generative AI across three dimensions: cognitive, affective, and behavioral dependency. Using a convenience sampling approach, data were collected from 1,295 university students with programming learning experiences involving generative AI. Item analysis was first conducted to examine item discrimination and reliability. Exploratory factor analysis was then employed to identify the underlying factor structure of the scale, followed by confirmatory factor analysis to test the model fit and construct validity. In addition, Pearson correlation analyses were performed to provide evidence of criterion-related validity. The results were generally consistent with a three-factor structure with satisfactory reliability and validity, indicating that the scale may serve as a useful instrument for assessing generative AI Dependency on programming among university students. Theoretically, this study extends the application of the ABC framework to AI-supported programming education and provides preliminary evidence that GenAI dependency may involve interrelated cognitive, affective, and behavioral components. Practically, this scale provides researchers and educators with a useful tool for investigating students' dependency on generative AI in programming and for informing instructional design and intervention strategies in programming education.

Key words: GenAI; dependency; programming learning; scale development; scale validation

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1. Introduction

In recent years, generative artificial intelligence (GenAI) has advanced rapidly, demonstrating significant application potential across multiple domains—including text generation, image creation, and data analysis [1,2]—and has gradually permeated various aspects of social production and daily life, such as healthcare [3–5], finance[6], creative design [7], and hospitality [8]. As its capabilities continue to improve, GenAI is exerting a profound impact on education [9,10], particularly transforming assessment and evaluation [11], teaching approaches [12] and learning paradigms [12] in higher education [13]. Within the context of programming education, GenAI—leveraging its natural language understanding, automatic code generation, and error detection and optimization capabilities—is increasingly emerging as a critical "programming assistance" tool, offering students immediate support during program design, debugging, and problem-solving processes [14–17].

AI programming support systems—represented by tools such as GitHub Copilot, as well as general-purpose generative AI assistants like ChatGPT—are capable of automatically generating corresponding code implementations based on brief comments, function signatures, or natural language descriptions [18–20]. By doing so, they potentially alleviate the cognitive burden that novice learners face during program conceptualization and implementation, enhance coding efficiency, and effectively mitigate common barriers encountered in the early stages of learning to program.

However, this high level of convenience has quietly given rise to generative AI dependence [21]. In educational technology and human–computer interaction research, dependence has been conceptualized, based on prior studies, as the sustained offloading of critical cognitive or decision-making functions onto an external system during task completion, resulting in functional reliance on technological support throughout the problem-solving process [22,23]. Programming education represents a particularly relevant context for examining GenAI dependency because learning to program involves not only code production but also the development of higher-order cognitive processes, such as problem analysis, algorithm design, debugging, logical reasoning, and code comprehension. In this context, dependency may occur when students shift from using GenAI as a supplementary learning aid to relying on it to perform core cognitive tasks. Such reliance may result in the outsourcing of essential mental processes that are central to programming competence and should be gradually developed through students' own thinking, reasoning, and practice.

In the absence of appropriate guidance, reliance on generative AI may give rise to multiple risks. Excessive dependence on GenAI can undermine students' cognitive capacities and impede their academic development [21]. This dependency pattern is further associated with cognitive inertia, whereby the repeated outsourcing of cognitive processes reduces intellectual engagement and autonomous thinking [24,25]. Moreover, sustained cognitive interaction with AI systems may trigger cognitive fatigue, a condition of mental exhaustion that impairs attention, reasoning, and judgment [26]. Excessive reliance on AI tools, such as ChatGPT, can lead to a range of adverse academic outcomes, including heightened academic passivity, increased exposure to misinformation, and diminished creativity and critical thinking capacities [27]. Overdependence on such technologies may further weaken students' engagement in independent analysis and reflective reasoning [10,28], while also undermining their ability to evaluate information critically and solve problems autonomously [29,30]. In addition, passive patterns of AI use—such as uncritical copy-pasting of generated content—are associated with surface-level learning approaches and limited understanding of complex concepts [31,32]. Collectively, these problematic dependency patterns on AI technologies pose substantial risks to educational outcomes [33]. These emerging concerns have already drawn attention from educational researchers and policymakers, highlighting the urgent need for scientifically grounded methods to identify and measure GenAI dependence among university students.

Although existing research has begun to pay attention to the phenomenon of dependence, many studies tend—often implicitly—to conflate “dependence” with “addiction” at the conceptual level, thereby overlooking their fundamental distinctions. In essence, AI dependence and AI addiction are markedly different. Dependence is not merely an observable usage behavior; rather, it constitutes a multifaceted psychological orientation encompassing cognitive, affective, and behavioral dimensions [34]. Cognitively, individuals perceive AI as a valuable tool for accomplishing specific learning, work-related, or problem-solving tasks. Affectively, their engagement with AI is primarily associated with feelings of efficiency and control, rather than being driven by emotional regulation needs. Behaviorally, dependence manifests as bounded and context-specific usage: AI is employed only within defined tasks or situations and does not spontaneously spill over into unrelated activities. Once the task is completed or the context changes, the use of AI naturally ceases—unlike addictive patterns, which persist across contexts in a repetitive, habitual, or automatic manner.

Although dependency and problematic use are conceptually related and may overlap, they are not identical constructs. In some measures of problematic technology use, dependency-related features are treated as one component within a broader problematic-use framework [35]. In the context of GenAI-supported education, problematic use may further involve issues such as plagiarism, academic dishonesty, inappropriate outsourcing of assignments, or other harmful and unethical uses of AI.

Although one study [36] has developed GenAI dependence scale, it performed only exploratory factor analysis without conducting subsequent confirmatory factor analysis to verify the factor structure. Moreover, it did not explicitly name or conceptually define the four extracted factors. Additionally, another notable limitation is the presence of identical items (item 7 and item 10), which raises concerns about redundancy and potential threats to construct validity. Though one study [34] developed a GenAI Dependence scale based on the ABC model structure, encompassing three dimensions—cognitive, affective, and behavioral—the instrument was constructed within a general usage context and has not been contextualized for programming-specific learning environments. Nevertheless, this framework provides a theoretically robust dimensional foundation for the present study.

However, it is concerning that some studies examining users' GenAI dependence have mistakenly employed measurement scales developed to assess addiction [27,37,38]. This conflation risks mischaracterizing adaptive or habitual use (e.g., relying on AI for productivity or decision support) as pathological addiction. Consequently, such methodological imprecision may inflate estimates of problematic AI use and obscure the nuanced psychological mechanisms underlying functional dependence versus compulsive, loss-of-control patterns typical of true behavioral addiction.

Therefore, developing a measurement instrument specifically designed to assess the degree of dependency on generative AI among university students in programming learning is not only theoretically necessary but also holds significant practical value in education. Accordingly, this study proposes and validates the Scale of Generative AI Dependency on Programming among University Students aiming to provide a conceptually clear, dimensionally sound, and psychometrically robust assessment framework that lays an empirical foundation for future instructional interventions, curriculum design, and educational policy-making.

2. Methods

2.1 Participants

This study adopted a convenience sampling approach. The questionnaire link was first shared with colleagues and friends through WeChat work groups and then forwarded by them to student WeChat or QQ groups under their supervision. The sample covered multiple universities across China (mainly located in Beijing, Hebei, Jilin, and other regions), with participants majoring in disciplines such as Computer Science and Technology, Software Engineering, Network and New Media, Mechatronic Engineering, Communication Engineering, Data Science, and Big Data Technology. The questionnaire was distributed online via the Wenjuanxing platform, with an average completion time of approximately 3–5 minutes. An electronic informed consent form was embedded on the first page of the questionnaire, providing detailed information on the research purpose, data usage, principles of anonymity and confidentiality, voluntary participation, and the right to withdraw at any time. The first question asked, “Are you willing to participate in this study?” Only respondents who selected “Willing to participate” were allowed to proceed with the questionnaire, while those who selected “Not willing to participate” were automatically exited. Selecting “Willing to participate” was regarded as confirmation that the participant had read and understood the informed consent and voluntarily agreed to participate in the study. Data collection was conducted from December 31, 2025 to January 3, 2026, during which students were generally still in the regular teaching period, with January 1 being the public holiday. However, because participants came from multiple regions and institutions, specific academic arrangements may have varied across universities. Out of 1,447 individuals, 152 declined to participate, resulting in a final sample of 1,295 participants. Random group assignment was conducted using SPSS. The sample was divided into two groups: Group 1 (n = 661) was used for Exploratory Factor Analysis (EFA), and Group 2 (n = 634) was used for Confirmatory Factor Analysis (CFA).

Table 1. Demographic information

Groups	Demographic Variable		N	%
EFA	Gender	Male	385	58.20%
		Female	276	41.80%
	Birthplace	Urban	171	25.90%
		Rural	490	74.10%
	Total		661	100.00%
CFA	Gender	Male	395	62.30%
		Female	239	37.70%
	Birthplace	Urban	148	23.30%
		Rural	486	76.70%
	Total		634	100.00%

2.2 Instruments

2.2.1 Scale of Generative AI Dependency on Programming among University Students

In the process of scale construction, this study draws on the theoretical framework of GenAI Dependency proposed by previous study [34], conceptualizing GenAI dependency as an ABC structural model consisting of Cognition, Affect, and Behavior. The ABC model was adopted because many established attitude-related scales have been developed based on this mature tripartite framework [39–41] In the context of GenAI Dependency on programming, this framework is useful for capturing students' emotional reliance on GenAI, their actual use-related dependency behaviors, and their cognitive evaluations of GenAI's capability and authority. Unlike prior scales that focus on general Dependency on broad usage contexts, the present study contextualizes this construct specifically within university students' programming learning environments.

During the item generation stage, a semi-structured interview guide was developed based on the ABC structural framework (see Appendix 1). Subsequently, five university instructors with extensive teaching and research experience in programming education were invited to participate in semi-structured interviews, lasting approximately 40 minutes. The interviews focused on students' dependency on GenAI in programming learning, including typical behavioral manifestations, representative scenarios, and underlying cognitive, affective, and behavioral mechanisms.

After transcription and organization of the interview data, two members of the research team independently conducted thematic coding. The coding process involved three steps. First, meaningful statements related to students' GenAI use in programming learning were identified from the interview transcripts. Second, these statements were grouped into preliminary thematic categories based on their semantic similarity and conceptual relevance. Third, the preliminary categories were compared with the ABC framework and further refined into item themes corresponding to the cognitive, affective, and behavioral dimensions. Conceptually similar and semantically overlapping categories were then merged and refined to ensure semantic clarity, avoid redundancy, and maintain consistency between item wording and dimensional definitions. Discrepancies between the two coders were discussed and resolved through consensus, and a third researcher was consulted when necessary. To further enhance the credibility and trustworthiness of the qualitative coding results, the coding results, preliminary thematic categories, and corresponding item themes were returned to the original five participating instructors for member checking.

This process resulted in 13 initial scale items, including 4 items for the Affect dimension (one sample item is *If generative AI does not respond for a long time during use, I feel uneasy and it affects my ability to continue programming*), 5 items for the Cognition dimension (one sample item is *I believe that the code generated by generative AI is more accurate than the code I write myself*), and 4 items for the Behavior dimension (one sample item is *When programming, even for tasks I can easily complete on my own, I still unconsciously turn to generative AI tools to generate code*).

Subsequently, the coding results and scale structure were returned to the original five participating instructors for member checking, in order to verify conceptual consistency, appropriateness of dimensional classification, and clarity of semantic expression. Then, a panel of 10 domain experts was invited to conduct an expert review of the scale items and dimension.

Experts evaluated the relevance and representativeness of each item within its assigned dimension using a four-point rating scale (1 = not relevant, 4 = highly relevant). Based on these ratings, expert content validity indices were calculated to systematically assess the content validity and structural validity of the scale. The expert evaluation results indicated that the ABC three-dimensional structure is theoretically sound and structurally coherent, with clearly defined dimensional boundaries and conceptual orientations. The full version of the scale is provided in Appendix 2 of the supplementary file.

2.2.2 Generative Artificial Intelligence Dependency Scale

To examine the criterion-related validity of the scale, the Generative Artificial Intelligence Dependency Scale [34] was adopted, and Pearson correlation coefficients were computed to assess the associations between Scale of Generative AI Dependency on Programming among University Students and the criterion scale. The Generative Artificial Intelligence Dependency Scale was developed based on the Media System Dependency framework and consists of three dimensions: affective dependency, behavioral dependency, and cognitive dependency, each comprising four items. One sample item for affective dependency is *I feel noticeably anxious or unsettled when I am unable to use Generative AI*. One sample item for behavioral dependency is *Even for tasks I am capable of completing independently, I give priority to using Generative AI to solve them*. One sample item for cognitive dependency is *When dealing with complex problems, my first thought is to seek ideas or answers from Generative AI*. The scale demonstrates excellent reliability, and empirical evidence indicates that it can effectively and stably capture students' levels of dependency on generative artificial intelligence.

2.3 Analytical procedure

Before conducting factor analyses, an item analysis was performed to examine the discriminative power and measurement quality of each item. First, the SPSS was applied to divide the EFA dataset into a high-score group and a low-score group based on total scores. Independent-samples t-tests were then conducted to compare item scores between the two groups, and items showing non-significant differences or weak discrimination were identified as candidates for removal. Second, item–total correlation coefficients were calculated to assess the consistency between each item and the overall scale structure; items with low correlations (e.g., < 0.40) were considered for deletion or revision. Finally, reliability analysis with item deletion (Cronbach's α if item deleted) was conducted to examine each item's contribution to internal consistency. Items whose removal led to a notable increase in Cronbach's α were regarded as contributing weakly to structural stability and were considered for exclusion.

Next, based on the retained items, an exploratory factor analysis (EFA) was conducted to examine the latent structure of the scale. Data suitability was first assessed using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity. When the KMO value exceeded 0.70 and Bartlett's test was significant ($p < 0.001$), the data were considered appropriate for factor analysis. Parallel Analysis using JASP [42] was used to determine the number of factors to be extracted. During item refinement, items were revised or removed according to the following criteria: low factor loadings, significant cross-loadings, and inconsistency between item and the predefined theoretical dimensions.

Based on the structure identified in EFA, confirmatory factor analysis was conducted with the CFA dataset using the CB-SEM module in SmartPLS [43]. SmartPLS was used as the analytical platform because it provides a covariance-based SEM (CB-SEM) module. In the present study, CFA was conducted using the CB-SEM approach rather than the traditional PLS-SEM algorithm. First, overall model fit was evaluated to assess the consistency between the proposed measurement model and the observed data using multiple goodness-of-fit indices, including the chi-square to degrees of freedom ratio (χ^2/df), CFI, TLI, RMSEA, and SRMR, with model adequacy indicated by $\chi^2/df < 3.0$, CFI and TLI > 0.90, RMSEA < 0.08, and SRMR < 0.08 [44,45]. Because the χ^2 statistic and its ratio to degrees of freedom are known to be sensitive to sample size, model fit was evaluated comprehensively based on multiple fit indices rather than relying solely on the χ^2/df criterion. Convergent validity was then examined using standardized factor loadings, Composite Reliability (CR), and Average Variance Extracted (AVE), with acceptable thresholds of loadings > 0.60, CR > 0.70, and AVE > 0.50 [46]. Discriminant validity was assessed using the Fornell–Larcker criterion, by comparing the square roots of AVEs with inter-construct correlation coefficients. Discriminant validity was supported when the square root of each construct's AVE exceeded its correlations with other constructs, indicating clear construct separation [47].

To examine the criterion-related validity of the scale, the Generative Artificial Intelligence Dependency Scale [34] was adopted as the criterion instrument. Pearson correlation coefficients were computed to assess the associations between the Scale of Generative AI Dependency on Programming among University Students and the criterion scale, in order to evaluate their convergence at the construct level and provide external validity support for the newly developed instrument.

2.4 Ethical considerations

The study protocol was reviewed and approved by the Science and Technology Department of Qinggong College, North China University of Science and Technology, which oversees research ethics compliance at the institutional level, and the ethical approval number is QGXYLL20250001. All research procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki and relevant institutional guidelines. An electronic informed consent form was presented on the first page of the questionnaire, outlining the study purpose, data use, anonymity and confidentiality, voluntary participation, and the right to withdraw. Participants were asked whether they were willing to participate, and only those who selected "Willing to participate" were permitted to proceed. This selection was regarded as confirmation of informed and voluntary consent.

3. Results

3.1 Item analysis

Before proceeding to factor analysis, item analysis was carried out to assess the discriminative power and psychometric quality of individual items. The EFA dataset was split into a high-score group (top 27%; total score ≥ 52) and a low-score group (bottom 27%; total score ≤ 39) according to total scale scores. Independent-samples t-tests were used to examine differences in item scores between the groups. All items showed statistically significant differences ($p < 0.05$), confirming good discriminative ability of the items.

The corrected item–remainder correlations ranged from 0.740 to 0.856, indicating that all items were strongly associated with the remaining scale score after excluding the target item from the total score (p = 0.01). It demonstrates strong internal consistency and good coherence with the overall construct.

Reliability analysis indicated that the overall internal consistency of the scale was excellent (Cronbach's α = 0.963). The results of the "Cronbach's α if item deleted" analysis showed that removal of any individual item did not lead to a significant increase in reliability, and therefore no items were deleted at this stage.

3.2 Exploratory factor analysis

An exploratory factor analysis was conducted on the items using EFA dataset to examine the underlying structure of the scale. Data suitability was first evaluated using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity. The KMO value was 0.923, indicating excellent sampling adequacy [48]. Bartlett's test of sphericity was highly significant, rejecting the null hypothesis that the correlation matrix is an identity matrix and confirming the presence of underlying factor structure.

Subsequently, parallel analysis with promax based on factor analysis supported the extraction of three factors, which was fully consistent with the predefined ABC theoretical model. As shown in Table 1, the first factor, Affect, demonstrated excellent internal consistency (Cronbach's α = 0.959) and explained 33.1% of the total variance. The second factor, Cognition, also showed excellent reliability (Cronbach's α = 0.955) and accounted for 33.0% of the variance. The third factor, Behavior, exhibited high reliability (Cronbach's α = 0.951) and explained 20.6% of the variance. Together, the three-factor structure provided strong empirical support for the theoretical ABC framework and demonstrated a clear, stable, and interpretable factor structure.

Table 2. EFA results

Item	Affect	Cognition	Behavior	Uniqueness	Communalities	Cronbach's α	Variance explained
A1	0.937			0.110	0.890	0.959	0.331
A2	1.039			0.056	0.944		
A3	0.870			0.145	0.855		
A4	0.818			0.120	0.880		
C1		0.822		0.207	0.793	0.955	0.330
C2		0.968		0.124	0.876		
C3		0.894		0.165	0.835		
C4		0.892		0.152	0.848		
C5		0.922		0.129	0.871		
B1			0.637	0.127	0.873	0.951	0.206
B2			0.656	0.150	0.850		
B3			0.630	0.173	0.827		
B4			0.846	0.071	0.929		

Note. Applied rotation method is promax. For clarity of interpretation, only factor loadings exceeding 0.40 were reported in the factor loading matrix.

3.3 Confirmatory factor analysis

To validate the factor structure and measurement model of the scale, confirmatory factor analysis (CFA) was then conducted on the CFA dataset using the CB-SEM module in SmartPLS. The CFA results indicated an overall acceptable to acceptable model fit. The chi-square test was significant, $\chi^2(62) = 257.36$, $p < .001$, with $\chi^2/df = 4.15$. Additional fit indices further supported the adequacy of the model: RMSEA = 0.070, 90% CI [0.062, 0.080], SRMR = 0.025, CFI = 0.981, and TLI = 0.976. Taken together, these indices indicate that the proposed measurement model demonstrates a good overall fit to the data.

3.4 Convergent validity

Convergent validity was assessed using standardized factor loadings, Composite Reliability (CR), and Average Variance Extracted (AVE). As shown in Figure 1, the standardized factor loadings ranged from 0.865 to 0.939, all exceeding the recommended threshold, indicating strong item–construct relationships. In addition, all three dimensions demonstrated excellent composite reliability, as shown in Table 2, with CR values of 0.961 for Affect, 0.957 for Behavior, and 0.959 for Cognition, all well above the recommended criterion of 0.70. Additionally, the AVE values were also high, with 0.862 for Affect, 0.847 for Behavior, and 0.824 for Cognition, exceeding the recommended threshold of 0.50. Together, these results provide strong evidence of good convergent validity for the measurement model.

Table 3. Convergent validity results

Dimension	Composite reliability (CR)	Average variance extracted (AVE)
Affect	0.961	0.862
Behavior	0.957	0.847
Cognition	0.959	0.824

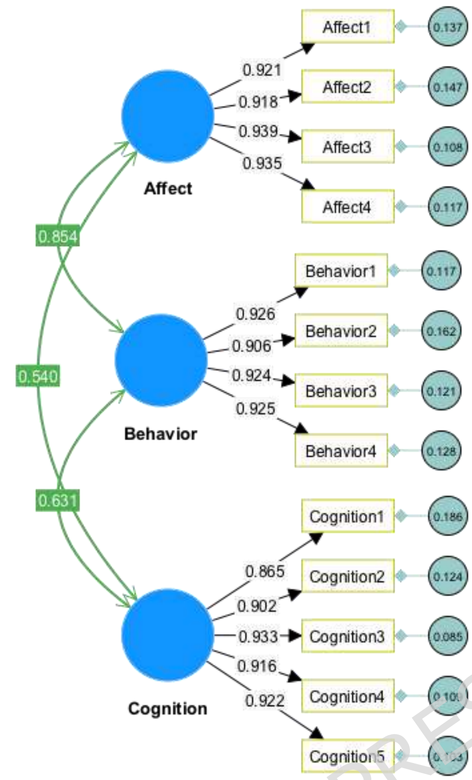

Figure 1. Standardized factor loading

3.5 Discriminant validity

Discriminant validity was assessed using the Fornell–Larcker criterion. As shown in Table 3, the square roots of the AVE values (diagonal elements) for each construct were 0.928 for Affect, 0.920 for Behavior, and 0.908 for Cognition, all of which were greater than the corresponding inter-construct correlations, providing strong evidence of adequate discriminant validity among the three constructs.

Table 4. Discriminant validity results

	Affect	Behavior	Cognition
Affect	0.928		
Behavior	0.854	0.920	
Cognition	0.540	0.631	0.908

3.6 Model comparison

To further examine the dimensionality of the GenAI Dependency on Programming Scale, an additional model comparison was conducted between the proposed three-factor model and a competing one-factor model. In the three-factor model, items were specified to load onto their respective affective, behavioral, and cognitive dependency factors, whereas in the one-factor model, all items were specified to load onto a single general GenAI dependency factor. The results showed that the three-factor model demonstrated better model fit than the one-factor model. Specifically, the three-factor model yielded more acceptable fit indices, including lower RMSEA and SRMR values and higher CFI and TLI values, with RMSEA = 0.070, 90% CI [0.062, 0.080], SRMR = 0.025, CFI = 0.981, and TLI = 0.976 for three-factor model and RMSEA = 0.292, 90% CI [0.284, 0.300], SRMR = 0.152, CFI = 0.663, and TLI = 0.596 for one-factor model. In addition, the information criteria further supported the superiority of the three-factor model, as it produced lower AIC and BIC values than the one-factor model with AIC=315.356 BIC=444.476 for three-factor model and AIC=3637.089, BIC=3752.842 for one-factor model. Since lower AIC and BIC values indicate a better balance between model fit and model parsimony, these findings provide additional evidence that the three-factor structure offers a more appropriate representation of the data than a purely unidimensional structure. Taken together, although the affective, behavioral, and cognitive dimensions were highly correlated, the model comparison results suggest that they should not be collapsed into a single undifferentiated factor. Rather, the three dimensions may be understood as theoretically distinguishable but empirically closely related facets of an overarching GenAI dependency construct.

3.7 Criterion-related validity

Criterion-related validity was examined by correlating the Scale of Generative AI Dependency on Programming among University Students with the Generative Artificial Intelligence Dependency Scale [34]. Pearson correlation results as shown in table 4 demonstrated that the dimensions and overall scores of the two scales were significantly and positively correlated, with correlation coefficients ranging from 0.464 to 0.734 (p=0.01). These results indicate moderate theoretical convergence between the two instruments and provide support for the criterion-related validity of the newly developed scale.

Table 5. Criterion-related validity results

Scale			1	2	3	4	5	6	7	8
Scale of Generative AI Dependency on Programming among University Students	1	Affect	--							
	2	Behavior	.817**	--						
	3	Cognition	.526**	.612**	--					
	4	Average score	.887**	.919**	.823**	--				
Generative Artificial Intelligence Dependency Scale	5	Affective	.706**	.672**	.514**	.718**	--			
	6	Behavioral	.683**	.669**	.464**	.688**	.847**	--		
	7	Cognitive	.667**	.663**	.510**	.698**	.847**	.921**	--	
	8	Average score	.717**	.699**	.519**	.734**	.940**	.965**	.964**	--

Note: **. Correlation is significant at the 0.01 level (2-tailed).

4. Discussion

The primary purpose of this study was to develop and validate the Scale of Generative AI Dependency on Programming among University Students, providing a psychometrically sound measurement instrument for assessing students' dependency on generative artificial intelligence within programming learning contexts. Grounded in the theoretical framework proposed by Jia et al. [34], this study adopted the ABC model of dependency (Affect, Behavior, and Cognition) and contextualized it within the domain of programming among university students, thereby developing a domain-specific instrument that captures the cognitive, emotional, and behavioral dynamics of generative AI Dependency on programming education. By situating generative AI Dependency on authentic programming learning scenarios, the scale reflects domain-relevant dependency patterns rather than generalized technology use, thereby enhancing its conceptual specificity and ecological validity.

To ensure rigorous scale development and validation, a multi-stage psychometric procedure was employed, including item analysis, exploratory factor analysis, confirmatory factor analysis, and Pearson correlation analysis for criterion-related validity. The EFA and CFA results consistently supported the hypothesized ABC three-dimensional structure. The present findings are also consistent with previous attitude-related studies based on the ABC model [39–41], which conceptualizes attitudes as comprising affective, behavioral, and cognitive components. In the context of AIGC, the three-dimensional structure identified in this study is broadly in line with Jia's findings [34], which also supported the applicability of the ABC framework. However, unlike previous research that examined general attitudes toward AIGC, the present study focuses specifically on students' dependency on generative AI in programming contexts, thereby extending the ABC model to a more domain-specific and behaviorally relevant educational scenario.

The items classified under the Cognition dimension reflect students' beliefs, judgments, and cognitive evaluations regarding the capabilities of generative AI in programming contexts. Specifically, items such as "I believe that the code generated by generative AI is more accurate than the code I write myself" and "I believe that generative AI is more effective than I am at identifying logical flaws or potential errors that I may overlook." reflect perceived cognitive superiority of generative AI, indicating a cognitive shift in authority attribution from the learner to the AI system.

The items classified under the Affect dimension consistently capture students' emotional reactions, affective dependence, and psychological responses associated with generative AI use in programming contexts. Specifically, items such as "If generative AI does not respond for a long time during use, I feel uneasy and it affects my ability to continue programming" and "Once the solution provided by generative AI contains flaws, I feel significant anxiety and uneasiness" reflect emotional reliance and anxiety-based dependence, indicating that students' emotional states are directly influenced by the availability and perceived reliability of generative AI.

The items classified under the Behavior dimension reflect students' actual usage patterns, habitual reliance, and behavioral substitution of generative AI in programming contexts. Items such as "When programming, even for tasks I can easily complete on my own, I still unconsciously turn to generative AI tools to generate code" and "When encountering programming problems, I first turn to generative AI rather than attempting to think independently or consult documentation." clearly indicate behavioral automaticity and habitual dependency, where AI use becomes a default behavioral response rather than a deliberate choice.

Furthermore, the scale exhibited strong convergent validity, discriminant validity, and criterion-related validity, indicating that the instrument possesses sound psychometric properties. Although the proposed GenAI Dependency on Programming Scale showed moderate to high correlations with the existing GenAI dependency scale, this finding should not be interpreted as evidence that the two instruments are redundant. Rather, the correlations indicate that the proposed scale shares a common conceptual foundation with general GenAI dependency measures while extending the construct into a more specific programming learning context. The incremental contribution of the present scale lies in its contextualized focus. Existing GenAI dependency measures are useful for assessing general dependency on AI tools, but they may not fully capture how dependency is expressed in programming-related learning activities. Programming involves distinctive cognitive and behavioral processes, including code generation, debugging, syntax correction, problem decomposition, algorithmic reasoning, code comprehension, and independent problem solving. Excessive reliance on GenAI in these processes may have unique implications for students' computational thinking, coding confidence, learning autonomy, and long-term programming competence. Together, these findings provide strong empirical support for the theoretical validity of the GenAI dependency framework in programming contexts and confirm that the proposed scale is a reliable and valid tool for measuring GenAI dependency among university students in programming contexts.

5. Limitations and suggestions

Despite the contributions of this study, several limitations should be acknowledged, which also provide directions for future research.

First, although the psychometric properties of the scale were systematically examined in terms of internal consistency, construct validity, and criterion-related validity, test–retest reliability was not assessed in the present study. As a result, the temporal stability of the scale could not be empirically verified. Future research should employ longitudinal or repeated-measures designs to evaluate the stability of the scale across time, thereby providing evidence for its test–retest reliability and strengthening its psychometric robustness.

Second, the data were collected primarily through self-report questionnaires, which may be subject to social desirability bias. This could lead participants to underreport their reliance on GenAI, especially if they perceive such dependency as academically or professionally undesirable. As a result, the measured levels of GenAI dependency may not fully reflect actual usage patterns or behavioral realities. Future studies could integrate multi-source data, such as behavioral logs, system usage records, learning analytics, and observational data, to provide more objective and ecologically valid measurements of generative AI dependency.

Third, although the target population of this scale is university students, the sample used in the present study was obtained through convenience sampling via WeChat and QQ groups and was mainly composed of students from rural backgrounds. This sampling approach may introduce selection bias and may not fully represent the diversity of the broader university student population, particularly students from different regions, institutional types, socioeconomic backgrounds, academic disciplines, and levels of programming experience. Therefore, the generalizability of the findings should be interpreted with caution, especially when applying the scale to more diverse groups of university students. Future research should further validate the scale using more balanced and representative samples, including university students from urban and rural backgrounds, different types of institutions, diverse majors, and varied programming learning experiences.

Fourth, because the present scale was developed based on the ABC framework, the item generation process was primarily organized around the cognitive, affective, and behavioral dimensions of GenAI dependency. This framework provided a clear theoretical structure for conceptualizing students' reliance on GenAI in programming learning and helped ensure coherence between item content and dimensional definitions. However, as the scale was developed within this predefined theoretical structure, it may not fully capture other potential manifestations of GenAI dependency that fall outside the ABC dimensions. Future research could build on the ABC framework while incorporating more inductive qualitative approaches to explore whether additional dimensions or context-specific features of GenAI dependency may emerge in programming or other AI-supported learning contexts.

Fifth, the initial item generation process did not directly involve interviews with university students, who are the target population of the scale. Although university instructors were invited as domain experts because of their extensive teaching experience and sustained observations of students' GenAI use in programming learning, their perspectives cannot fully substitute for students' first-person experiences. This may limit the depth to which the initial item pool captured students' subjective perceptions of GenAI dependency. Future studies could invite university students to participate in the further enrichment, refinement, and evaluation of the scale items through student interviews, cognitive interviews, or focus group discussions. Such efforts would help expand the item pool, improve item wording, and further strengthen the content validity and theoretical grounding of the scale.

A further limitation is the potential presence of common method bias. Because the data were collected using self-reported questionnaires at a single time point, shared method variance may have partly contributed to the high correlations among the dimensions. Although additional model comparisons indicated that the three-factor structure was not fully attributable to a single common factor, future studies should use multi-source data, longitudinal designs, or objective behavioral measures to further examine and reduce possible common method bias.

Finally, although the scale demonstrated very high internal consistency, the alpha coefficients and inter-item correlations should be interpreted with caution. Very high alpha values may indicate that some items are closely related in wording or content coverage, rather than simply reflecting stronger reliability. In the present study, the items were not directly repetitive in content, but some dimensions shared similar phrasing patterns and contextual framing, which may have contributed to the high inter-item correlations. Therefore, future studies should further examine potential item redundancy and consider refining or rewording items with similar phrasing or contextual framing. Any revised or shortened version of the scale should be revalidated to ensure that it retains adequate reliability, validity, and theoretical coverage.

6. Implications

This study provides useful implications for research and practice in AI-supported education by offering a theoretically grounded measurement framework that is empirically consistent with the proposed conceptualization of GenAI Dependency on programming education.

From a theoretical perspective, the proposed ABC structure offers a useful framework for understanding GenAI Dependency on programming education. Rather than providing definitive evidence for fully distinct dimensions, the findings suggest that GenAI dependency may involve closely related affective, behavioral, and cognitive components, including cognitive trust, emotional reliance, and behavioral substitution. This interpretation contributes to ongoing discussions on human–AI interaction and technology Dependency on AI-supported education. This extends dependency theory into AI-mediated learning environments and indicates that the ABC framework may inform future research on GenAI Dependency on other domains, such as writing, design, engineering, and mathematics. However, such applications should be carefully contextualized to specific task settings and supported by further empirical validation. Meanwhile, the present scale extends existing measures by offering a programming-specific assessment of GenAI dependency. It enables researchers and educators to examine not only whether students depend on GenAI, but also how such dependency is manifested affectively, behaviorally, and cognitively within programming learning. In this sense, the scale provides incremental value by offering a more fine-grained and pedagogically relevant instrument for studying GenAI Dependency on programming education.

From a practical perspective, the scale provides educators and institutions with a diagnostic and intervention-oriented tool for identifying different forms of GenAI Dependency on programming education. In the Chinese higher education context, the rapid integration of generative AI tools into programming education presents unique opportunities and challenges. On the one hand, AI-assisted programming tools can enhance learning efficiency and provide timely support for students. On the other hand, the strong emphasis on academic performance and task completion may increase the likelihood that students rely on AI-generated solutions without fully engaging in the underlying problem-solving process. Therefore, monitoring and understanding students' dependency on generative AI is particularly important for promoting the balanced development of programming competence, critical thinking, and independent learning skills.

7. Conclusion

This study developed the Scale of Generative AI Dependency on Programming among University Students based on the ABC framework (Affect, Behavior, Cognition). The findings provide initial support for the scale's reliability and validity. The instrument may serve as a useful tool for future research and practical assessment in AI-supported learning contexts.

Ethical Approval and Consent to Participate

This study was reviewed and approved by Qinggong College, North China University of Science and Technology, and the ethical approval number is QGXYLL20250001. All research procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki and relevant institutional guidelines. An electronic informed consent form was presented on the first page of the questionnaire, outlining the study purpose, data use, anonymity and confidentiality, voluntary participation, and the right to withdraw. Participants were asked whether they were willing to participate, and only those who selected "Willing to participate" were permitted to proceed. This selection was regarded as confirmation of informed and voluntary consent.

Competing interests

The authors declare that they have no competing interests related to this study. All authors have disclosed any potential conflicts of interest.

Author Contributions

Hao Zhang: Conceptualization, supervision, revision.

Yanchao Yang: Methodology, validation, data analysis, writing—original draft.

Ying Zhang: Instrument development, data curation, writing—review and editing.

Baishuang Qiu: Data collection, formal analysis, visualization.

Jiongzhao Yang: Literature review, methodological support, writing—review and editing.

Data Availability Statement

The data supporting the findings of this study have been uploaded and are available for access.

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Consent for Publication

Not Applicable

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